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Low-Resource Mispronunciation Detection and Diagnosis: Applications to Irish

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Abstract

Minority languages are often left out of modern NLP developments. Though these technologies can provide platforms to connect discontiguous communities of language speakers, the dominance of a select few languages in the digital realm reveals a wide disparity in resources available to smaller languages in data-intensive applications such as Computer-Assisted Pronunciation Training (CAPT) and Mispronunciation Detection and Diagnosis (MDD). To work around the data limitations inherent in low-resource MDD relevant for building CAPT systems for minority languages such as Irish, we propose two possible strategies: by confidence-based ensembles of ASR models leveraging comparatively plentiful monolingual data for Irish and English, as the most common Irish learner first language (L1); as well as by synthesis of scarce phonetically transcribed English L1 Irish second language (L2) speaker data by means of transcribing Irish speech with an English ASR model, and synthesizing speech from these transcriptions with comparatively robust English TTS systems. Results showed however that no system beat the monolingual Irish baseline at detecting mispronunciations, but illuminate failure points with respect to cross-corpus comparability that could indicate promising ways forward for applications facing similar data constraints as the current work.

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Acronyms

AI	Artificial Intelligence 9
ASR	automatic speech recognition 2, 3, 8, 9, 13–22, 26–28, 35–37, 40, 41
BERT	Bidirectional Encoder Representations from Transformers 17
CALL	Computer-Assisted Language Learning 3, 9, 13, 14
CAPT	Computer-Assisted Pronunciation Training 2, 3, 8, 9, 13–15, 40
CD	Correct Diagnostic 28
CER	character error rate 22, 23, 31
CNN	Convolutional Neural Network 17
CTC	Connectionist Temporal Classification 17, 36
DE	Diagnostic Error 28
DER	Diagnostic Error Rate 28, 31
DNN	deep neural network 19
DPAD	diversified and preference-aware decoding 20, 35, 41
E2E	end-to-end 15–17, 19
FA	False Acceptance 27, 28
FAR	False Acceptance Rate 28, 31
FLEURS	Few-shot Learning Evaluation of Universal Representations of Speech 27, 43
FR	False Rejection 28
FRR	False Rejection Rate 28
G2P	grapheme-to-phoneme 22, 24, 26, 27, 36
GCS	Google Cloud Service 27
GMM	Gaussian Mixture Model 16
GOP	goodness of pronunciation 15
HMM	Hidden Markov model 15, 16, 19
IPA	the International Phonetic Alphabet 12, 15, 22, 24–27, 37
L1	first language 2, 8, 11, 19, 20, 26, 41, 42
L2	second language 2, 8, 11, 12, 16, 18–20, 22, 27, 37, 38, 41
LM	Language Model 41
LR	Logistic Regression 25, 27
MD	Mispronunciation Detection 9, 15, 16, 19, 20, 22, 26
MDD	Mispronunciation Detection and Diagnosis 2, 3, 8, 9, 14, 18–20, 22, 24, 27–29, 38, 40, 41
MFA	Montreal Forced Aligner 27, 36
ML	Machine Learning 8, 16–19
MOOC	massive open online course 13
MT	Machine Translation 13, 20
NLG	Natural Language Generation 20
NLP	Natural Language Processing 2, 3, 9, 10, 12, 13
NN	neural network 10, 12, 20
OOV	out of vocabulary 27
P2P	phoneme-to-phoneme 20
ROVER	Recognizer Output Voting Error Reduction 18, 37
Seq2Seq	sequence-to-sequence 20, 26
SSML	Speech Synthesis Markup Language 26

TA	True Acceptance 28
TL	Transfer Learning 13
TR	True Rejection 28
TTS	text-to-speech 2, 8, 13, 18, 20, 22, 26, 35
WALS	the World Atlas of Linguistic Structure 9
WER	word error rate 19, 20

1 Introduction

Language competency is crucial for forming identity, securing work opportunities, and navigating society at large. As our surroundings become ever more digitally interconnected, how we interact with it is increasingly mediated by language technology, for both good and ill. Such technologies are often identified as promising tools to promote language use and language learning, but are also implicated in the coalescence of online interaction around a few dominant languages such as English, complicating the glowing promises often offered by their proponents. Low-resource language communities open to language technologies are often left out of most recent technological advancements given the relative lack of resources at their disposal, necessitating approaches tailored to these limitations if the promise of cutting-edge language technologies is to be realized equitably. The momentum of the pre-training/fine-tuning paradigm within automatic speech recognition (ASR) and Machine Learning (ML) in recent years offers a glimmer of hope to those working toward making accessible tools for such communities, enabling the use of more plentiful data to support tools for languages facing various levels of resource scarcity.

Promoting speech technologies for language learning could be of particular benefit for languages such as Irish which endeavor to propagate native models of speech to motivated learners outside of traditionally Irish-speaking areas. Through CAPT applications built with careful use of existing data, we might find the scaleable, resource-conscious ways to support such communities and narrow the gap between tools built for digital behemoths such as English, and those built for languages without similar access to digital resources. Though Irish is a language with several distinct dialects, we choose here to focus our efforts on Ulster Irish as our object of study as a simplifying measure, leaving more representative coverage to future work in this domain.

To this end we explore two potential avenues to overcoming the data limitations faced by Irish and other languages like it with respect to learner data: one, by using the data we *do* have by harnessing readily available monolingual data in a model ensemble; and the other, by imitating the data we *wish* we had with text-to-speech (TTS)-generated learner approximations for training a combined model. These approaches are formulated as the following research questions:

1. To what extent does a resource-conscious ensemble of monolingual ASR models (Irish and English) offer improved discrimination of learner errors compared to a monolingual Irish model?

2. To what extent does a model trained on TTS-generated synthetic mispronunciation data offer improved discrimination of learner errors compared to a monolingual Irish model?

The unifying goal of these approaches is exploring the feasibility of overcoming the scarcity of phonetically annotated L2 learner data for MDD applications. One approaches the problem with ensembles of monolingual ASR models, leveraging more plentiful L1 speaker data, and another using a schema of data generation by leveraging robust TTS systems to approximate learner speech: a potential low-cost alternative to large-scale phonetic annotation. Through this, we hope to illuminate possible ways forward for low-resource language communities interested in developing automated technologies for language learning and pronunciation training.

2 Background

In the following section we will outline the context motivating our current work within Natural Language Processing (NLP) and ASR for minority languages. We begin with the state of such languages in current research, highlighting some promising trends as well as some thusfar recalcitrant problems hindering more equal access to the rapid advances in language technology. We then proceed to outline the general research space of the current work: Computer-Assisted Language Learning (CALL) and CAPT before digging into the core task of CAPT systems: Mispronunciation Detection (MD) and MDD. We conclude with a technical overview of modern ASR systems used for such tasks, leaving the specific efforts of researchers to overcome data limitation inherent to low-resource languages in chapter Chapter 3.

2.1 Irish & The Predicament of Minority Languages in Natural Language Processing (NLP)

Developing language technologies that can scale beyond the language they are designed for is no new goal for NLP research. The value of such a property is apparent: transferring an existing system seamlessly to another language could potentially save significant resources for language communities without the ability to fund such development themselves. Actually achieving this goal in practice, however, is no simple endeavor, typically requiring some level of linguistic awareness which is all-to-often lamentably absent (see Bender, 2011; Hedderich et al., 2021; Joshi et al., 2021, inter alia). For example, success in applying word-based n-gram approaches to a language rests on the language's level of inflectional morphology: the lower the better. Higher morphological complexity together with variations in word order raise data sparsity problems which n-gram approaches rooted in English struggle to handle (Bender, 2011). This should come as no surprise, given the relatively fixed word order and low levels of inflectional morphology present in English, but it illustrates a need for caution: systems developed for a given language may make implicit assumptions about language structure which do not generalize well. Linguistic typology can provide important clues as to what features are shared between the original development language and possible languages of extension for a system. Information of this kind has, for many of the worlds languages, already been gathered by linguists. Perhaps the most renowned database of typological information is the World Atlas of Linguistic Structure (WALS) (Dryer et al., 2024), a free, online resource currently boasting 152 chapters with detailed descriptions of 192 linguistic features spanning over 2,600 of the world's languages at the time of writing. By explicitly mobilizing linguistic knowledge already painstakingly gathered by linguistic typologists, we can identify where languages agree, where they differ, and hopefully identify some of the implicit, ungeneralizable assumptions that have hindered past efforts to create more accessible language technologies.

Despite claims of language-agnostic systems often touted by proponents of emerging Artificial Intelligence (AI) systems, these have often also fallen well short of such promise (Bender, 2011). The overwhelming majority of the world's languages still have no footprint in emerging language technologies (Joshi et al., 2021). In the past,

building neural network (NN)-based language technologies demanded immense quantities of labeled data: a high bar of entry to language communities with limited if any access to such resources, and an ongoing issue which continues to stimulate a body of research dedicated to overcoming it (see Magueresse et al., 2020, for an overview). For languages where data availability is no obstacle, research and development can proceed unfettered by the prohibitive cost of curating datasets from scratch. As our daily lives grow increasingly integrated with the digital realm, language communities without the same support are obliged to switch to more digitally dominant languages (often English) to gain access to these new resources, narrowing the opportunities to engage with resources and services through the medium of their community language (Ní Chasaide et al., 2019). A particularly sobering taxonomy illustrating the states of languages facing such disparities is formulated by Joshi et al. (2021) and reproduced in Table 2.1, which outlines the challenges faced by languages in resource terms in the digital space, and how dominant a small group of languages are within it. Those at the bottom of the data availability scale at level 0, 'The Left-Behinds' have virtually no data of any kind available to support language technology development. Those in the middle at level 2 or 3 have some resources, net presence, and/or research communities supporting them, but critically lack in substantial quantities of the labeled data typically required for cutting-edge NLP tools. At the top, languages like English reap the lion's share of overall investment and development. Of particular note for the privileged few that find themselves at the top of the heap is their typological similarity, being drawn as they are from a few dominant language families (and even dominant branches within these larger families). This state of affairs constitutes a sort of typological echo-chamber for the cutting edge of NLP developments, a point which we will return to shortly.

Class Descriptions	**Example Languages**	**% of total languages**
0 The Left-Behinds: Virtually ignored in language technology. Exceptionally limited resources available, even with respect to unlabeled data.	Dahalo, Bora	88.17%
1 The Scraping-Bys: Some unlabeled data. With organized promotion and data collection, there is hope for improvement in coming years.	Fijian, Navajo	8.93%
2 The Hopefuls: Limited labeled data. Support communities help these languages survive, and there is promise for NLP tools in the near term.	Zulu, **Irish**	0.76%
3 The Rising Stars: Strong web presence and thriving community online. Lacking labeled data. Good potential for NLP tool development for these languages.	Indonesian, Hebrew	1.13%
4 The Underdogs: Much unlabeled data, and less but still significant labeled data. Dedicated investment from NLP communities.	Russian, Dutch	0.72%
5 The Winners: Dominant online presence with massive investment and resources.	English, German	0.28%

Table 2.1: Data availability & status taxonomy of languages adapted from work by Joshi et al. (2021).

Despite some advantages not afforded other minority languages, Irish still finds itself struggling to maintain a footing in the the digital realm, placed by Joshi et al. (2021) in class 2 of the taxonomy in Table 2.1. It enjoys ongoing investment by the Irish state, nominal status of Irish as the first national language of the Republic of Ireland,

and research dedicated to its promotion (e.g. through the ABAIR initiative dedicated to developing speech technologies for Irish, see Chasaide et al., 2017). At the same time, it is a typological outlier in several respects: it is a verb-initial language with relatively complex inflectional morphology, and a distinct (though still Latin-based) orthography. Features like these put Irish at odds with many languages in the high-resource echo-chamber, complicating the ability to leverage cutting-edge models to linguistic features with no representation in a model's training data. Furthermore, though we have treated Irish as a single entity thus far, an important complication reveals itself in the discontiguous nature of the Irish-speaking areas, referred to as *the Gaeltacht* or *the Gaeltachtaí*(pl.). Each of the three main areas (i.e. Ulster, Connacht, and Munster 2.1) speak markedly distinct varieties of Irish, necessitating labeled data from each variant if these groups are to be adequately serviced by new technologies (Ní Chasaide et al., 2019). In the face of such limitations, today's data-hungry tools simply cannot be expected to perform to the same level on languages like Irish without the necessary resources. It should be noted that the pretraining/finetuning paradigm of recent massive multilingual models does mitigate this demand for data somewhat by leveraging unlabeled cross-lingual data, reducing the need for labeled data in the language being finetuned to (Hedderich et al., 2021; Joshi et al., 2021; Ranathunga et al., 2021), but for other languages without any labeled data to their name, this is a small comfort.

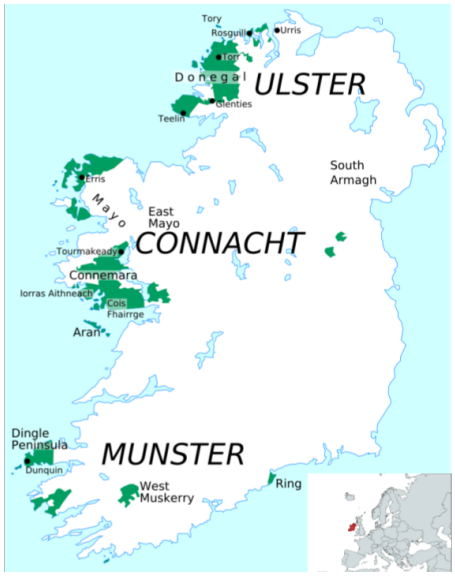


Figure 2.1: Area of the Gaeltachtaí colored in green. Originally uploaded by user Mahagaja and downloaded from Wikimedia Commons. Highlighted european map was created with mapchart.net

Though the hurdles facing the Irish language in the digital sphere are a relatively recent concern, the issues facing it in the real world are anything but. It is currently classified by UNESCO as being *definitely endangered* (Moseley and Nicolas, 2010), following centuries of varying rates of contraction due to encroachment by English. Irish survives as a community languages in the aforementioned Gaeltacht areas, though even there it is estimated that only 24% of inhabitants speak Irish on a daily basis (Ní Chasaide et al., 2015). Despite the state of Irish as a L1, it is comparatively strong as a L2 (Broin, 2014). A growing number of parents seek Irish-medium education for their children outside the Gaeltacht, and immersive summer courses remain popolar among adults looking to learn or reconnect with the language. This encouraging L2 engagement intersects with thorny issues of supply, however, as many of the teach-

ers are not themselves native speakers with an accompanying native grasp of the structure and sound of the language (Ní Chasaide et al., 2015, 2019).

This limited native speaker model for L2 speakers is problematic for teaching pronunciation, particularly for the acquisition of some sound contrasts critical to disambiguating the Irish grammar. Perhaps chiefly among these, contrasts between the secondary articulation of consonants into *palatalised* and *velarised* variants play an instrumental role in a number of grammatical functions, such as in the formation of certain plurals and genitive marking (Broin, 2014; Gabriele, 2018; Snesareva, 2016; Stenson, 2020). Since the Roman alphabet does not provide symbols for this distinction, Irish orthography marks it via adjacent letters as seen in row (b) of table 2.2: so-called *slender* vowels ('i' and 'e') around a consonant denote *palatalisation*, while *broad* vowels ('a', 'o', and 'u') mark *velarisation* (Stenson, 2020).[1] Mutation effects are another pervasive element of the Irish grammar relying on sound alterations, the most common of which are *lenition* and *eclipsis*. Lenition, traditionally termed *séimhiú* (/ˈʃeːvʲuː/), is commonly marked with an 'h' following the lenited consonant as seen on Table 2.2, and originally denoted a weakening in the manner of articulation, though the relationship between consonants and their lenited versions is less immediately apparent now for some consonants (Stenson, 2020). Eclipsis, traditionally *úru* (/ˈʊɾˠuː/), involves replacing the original consonant with a nasalized or voiced version, and is denoted by appending the new sound character before the consonant being eclipsed2.2. These and other structural underpinnings of the language can have far-reaching implications for intelligability if not adequately mastered by students. Indeed, a study undertaken by Broin (2014) reveals that realisations of phonological details like those above have error rates of over 50% on average for urban speakers, with some as high as 82%—a stark departure from their gaeltacht counterparts which lie below 10%. Providing wider access to better native models of pronunciation—not to mention native models of morphology—could do much to close this gap, making mutual intelligability between gaeltacht and urban speakers more attainable.

Table 2.2: Examples use of a. consonant velarisation & b. palatalisation c. séimhiú & d. úru. Adapted from Ní Chasaide et al. (2015) with determining orthography in bold.

	Orthographic	the International Phonetic Alphabet (IPA)	Translation
a.	b**ád**	/bˠaːdˠ/	'boat'
b.	bá**id**	/bˠaːdʲ/	'boats'
c.	do **bh**ád	/dˠɔ waːdˠ/	'your boat'
d.	ár **mb**ád	/ɛɾʲ mˠaːdˠ/	'our boat'

2.2 From Text to Speech: Approaches to Low-Resource Scenarios

With the rise of NN-based language technologies, the data-hungry nature of such tools underscores the urgency of addressing the kind of resource disparities outlined above. Making such tools accessible to languages without the same strong data foundations as English is an active area of ongoing research, though even within popular languages such as English, non-standard domains and tasks types can constitute low-resource areas which lack suitable quantities of training data. These data disparities can be categorized along several dimensions, such as those proposed for NLP by Hedderich et al. (2021) as: availability of *task-specific labeled data* for the target language

[1] For initial consonants, the following vowel plays the determining role, for final consonants, the preceding vowel, and for medial consonants, the vowels flanking it.

or domain, availability of *unlabeled* language- or domain-specific data, or the availability of *auxiliary* data. This latter kind of data is diverse, as it is data not directly labeled for the task at hand, but which can still be indirectly useful, from labels specific to another language/domain, to knowledge bases such as entity lists, or automated labels from Machine Translation (MT) systems (Hedderich et al., 2021).

To address the different dimensions of resource scarcity outlined above, various approaches have been developed which Hedderich et al. (2021) splits broadly into those which *generate additional labeled data*, and those employing *Transfer Learning (TL)*. Faced with limited gold-standard annotated data, researchers employ strategies of the former type to automatically produce labeled alternatives. These strategies can themselves be broadly grouped as *data augmentation*, where task-specific labeled data is used to make more labeled data, such as with Back-Translation for MT where a target-to-source translation model is used to obtain a synthetic parallel corpus from a monolingual target corpus (Ranathunga et al., 2021), and *distant supervision* which produces labels for existing unlabeled data, for example in Cross-Lingual Annotation Projection where a task-specific classifier is trained for a high-resource language, then projected onto text from a low-resource language using a paralell corpus. For TL, in contrast, instead of creating or extending task-specific training data, the focus lies on reducing the need for such data by leveraging models or learned representations from other languages/domains. This approach has been particularly successful in in recent years with the advent of models like BERT (Devlin et al., 2019) and Wav2Vec2 (Baevski, Zhou, et al., 2020) which are *pre-trained* on vast quantities of unlabeled data to then be *fine-tuned* for specific downstream tasks. This pretraining/fine-tuning paradigm can be particularly advantageous for languages or domains where labeled data is limited.

Although these strategies are commonly employed in NLP, analogous trends can be found in computer vision as well as speech to tackle similar limitations in data. TL and the aforementioned pretraining/fine-tuning paradigm has made strides with self-supervised models like Wav2vec2, outperforming previous state-of-the-art ASR models with 100 times less data, starkly reducing the demand for labeled speech (Baevski, Zhou, et al., 2020). Alongside TL, ensemble methods have also emerged as a promising tool for low-resource contexts. Here, specialist models with complementary attributes can be combined to perform better than any one of its constituents for novel tasks (Arunkumar et al., 2022; L. Deng and Platt, 2014; Fiscus, 1997; Gitman et al., 2023, inter alia). Various methods of data augmentation and data synthesis are also prevalent to improve ASR performance for low-resource languages, including voice transformations, where noise or other alterations are introduced to recordings to extend existing data, or generating synthetic audio data samples with a TTS system to bolster training data when authentic speech corpora are lacking (Bartelds et al., 2023; Zhang et al., 2022).

2.3 Computer-Assisted Language Learning (CALL) & Computer-Assisted Pronunciation Training (CAPT)

Despite efforts to the contrary, the rise of digital technologies is often implicated in the acceleration of already precpitous rates of decline for endangered languages (Ní Chasaide et al., 2015, 2019, inter alia). However, it is a trend that cuts both ways, as the same technologies that squeeze certain languages out of the digital realm are also making space for communities of language learners to come together towards their common goal through CALL platforms and massive open online course (MOOC)s. The increased presense of technology both in and outside the classroom brings with

it broad implications for traditional pedgagogy, enabling more autonomous and flexible modes of learning for students (Chapelle, 2008) particularly for learners looking to autonomously improve their pronunciation via CAPT systems. This technological shift has increased the financial viability of courses for endagered or otherwise less commonly taught languages, allowing teachers to draw from a more geographically dispersed enrollment pool and provide courses otherwise impossible to offer. Despite this potential, tension between the technology and pedagogy underpinning CALL and CAPT systems remains a well-documented issue, and their increasing prevalence in the modern language learning landscape has renewed calls for greater collaboration between pedagogical and technical experts when designing these systems (Rogerson-Revell, 2021). This divide between technical experts and real-world application also crosscuts the aforementioned struggles of low-resource language communities in making the most of recent technological developments, though some languages such as Irish are still fortunate enough to have research groups working to bridge that gap (Ní Chiaráin, 2022; Ní Chiaráin et al., 2022).

The proliferation of CALL software for self-study such as Duolingo, Babbel, and Rosetta Stone has renewed interest in the role of immediate and personalised feeback for pronunciation training in CAPT systems. More recent research indicates that language learners may need explicit and targeted feedback on their pronunciation in order to improve (Bajorek, 2017), despite the long-held understanding that error correction typically does not meaningfully influence acquisition (Krashen, 1984). Perhaps as a consequence of this established view, explicit pronunciation training has been notably absent from language classrooms in recent decades, and many CALL platforms carry on this legacy with binary right-or-wrong feedback mechanisms that do not make use of the potential for more effective, targeted feedback (Bajorek, 2017). The feedback is also frequently unexplained, making such binary judgements about pronunciation quality opaque to the student and thus more difficult to act upon. The ideal individualized, explaination enriched feedback would normally carry a steep price tag for the student of an individual tutor, say, but CAPT systems can potentially lower this barrier of entry considerably by automating the same kind of undivided feedback in a one-to-many form scalable to many geographically disperesed students at once.

Although the scope of this work does not evaluate the effectivness of any particular pedagogical application, we recognize the potential of CAPT systems and the need for pedagogical and linguistic awareness in their design to more fully deliver on their promise. To that end, we endeavor in this work to contribute with linguistically aware ASR strategies which can hopefully help bridge the gap between cutting-edge language technology and the needs of minority language communities, supporting pedagogically sound feedback to learners.

2.4 Pronunciation & Mispronunciation Detection and Diagnosis (MDD)

To realize the often untapped potential of CAPT and lay the groundwork for actionable feedback, the CAPT system must be able to identify deviations in student pronunciations from the target pronunciation and determine how it differs in ways interpretable to the student. To do this, we can start by specifying what we mean by pronunciation. In broad strokes, the human speech apparatus may be seen as a collection of subsystems which can emit signals over parallel channels (Engstrand, 2004, p. 173). Although speech is a continuous signal, we can map symbols of discrete speech sounds–phones–onto subsegments of this signal. These discrete units—the vowels and

consonants that constitute words—are *segmental* features of speech. The prosody, intonation, stress, and other such elements of speech can be seen as superimposed on these segmental features, and are thus termed *suprasegmental* features of speech (Engstrand, 2004). Human cultures have an array of strategies for representing speech in written form, ranging from logographic writing systems like Chinese Hanzi where one symbol represents one word, to different varieties of sound-based systems such as syllabic for Japanese hiragana or katakana, alphabetic like the Latin alphabet used here, or consonantal as in Semitic scripts (Jurafsky and Martin, 2025). Many, though by no means all, approaches to MD within CAPT focus on recognizing these segmental features in speaker recordings and comparing their phonetic transcriptions to a canonical pronunciation of speech for the target word (Korzekwa et al., 2022). For our purposes, we will similarly narrow our investigation of MD to first focus to detection of these segmental features, representing pronunciation as strings of phones using the IPA standard of phonetic notation. Though suprasegmental features can also play a crucial role in disambiguating meaning, investigating them is beyond the scope of this thesis.

Segmental recognition for MD can be broadly divided into two approaches: those which align speech signals to canonical segment sequences, and those which first extract segments—generally phones or phonemes—from speech, then aligning the extracted sequence of segments to a canonical sequence for comparison (Korzekwa et al., 2022). The process of aligning speech to text which characterizes the former approach is termed *Forced Alignment*. Classic pronunciation scoring generally focused on these Forced Alignment approaches, which typically relied on log-likelihood scores and log-posterior scores derived from Hidden Markov model (HMM)-based ASR model outputs (S. M. Witt, n.d.). The latter of these scores soon became the de facto standard due to its higher correlation with human assessments. Building further on this development, S. M. Witt (2000) introduced the widely used goodness of pronunciation (GOP) measure of pronunciation quality. The resultant score can be compared against a threshold value to determine how well the speaker's pronunciation matches the canonical model pronunciation (S. Witt and Young, 2014). The classic formulation of GOPs base form was given by S. M. Witt (2000) as the log of the posterior probability of any phone given an acoustic segment normalized by the length of this segment as given below by:

$$GOP(q_i) \equiv \log(P(q_i|O))/NF(O) \tag{2.1}$$

where O is the speech segment, and $NF(O)$ is the number of frames of segment O.

Although GOP and other posterior-based scores continue to be used for MD with modern ASR architectures (see Gong et al., 2022; Parikh et al., 2025, inter alia for examples), with the advent of conceptually simpler transformer-based end-to-end (E2E) models in the last decade, researchers have also pursued free-phone output sequence alignment to detect mispronunciation (Wu et al., 2021). This alternative circumvents the need for phone-level forced alignment of speech signals to reference transcriptions (Leung et al., 2019; Lo et al., 2020), and can be used to diagnose mispronunciation errors instead of simply detecting them by stopping at scoring pronunciation quality.

To detect mispronunciations at the segmental level, our approach starts with the former family of approaches outlined above: using forced alignment-derived phones as an intermediate step to extract phones. This focus on articulatorily meaningful units of speech could lay the ground for more informative and actionable feedback in any future CAPT extension, particularly in comparison to binary assessment strategies of simply identifying correct or incorrect pronunciation, but this focus introduces its own set of challenges. Recognizing phones requires fine-grained annotations at

that level, which are difficult to obtain, particularly for L2 speech (Zhang et al., 2022). This framing of the MD problem is a data-scarce task to pursue, but the modern innovations in ASR architectures leveraging the aforementioned pre-training/fine-tuning paradigm could make it an increasingly feasible one, even for low-resource languages.

2.5 Speech and ASR

The critical technology underpinning segmental MD approaches is the ASR system transcribing spoken words into writing. ASR research spans many decades, having matured from its origins as a barely useful method of interface to the point where speech is the primary modality of interaction for popular forms of human-machine communication today (Yu and L. Deng, 2015). Some simpler tasks have been long since solved, such as simple yes-no recognition or digit recogniton (i.e. 0-9). Transcribing strings of phones, phonemes, or words like today's ASR models however, is far more complex, with many tens of thousands of possible vocabulary items to predict (Jurafsky and Martin, 2025). In recent decades, however, even these tasks have proven tractable under controlled conditions, as evidenced by the popularity of different commercial voice-controlled digital assistants such as Amazon's Alexa,[2] Apple's Siri,[3] or Google Assistant.[4] Modern ASR need to be robust to more than just vocabulary size, however. Jurafsky and Martin (2025) highlight other consequential dimensions of speech variability such as who the speaker is talking to; *read speech* such as that produced when humans talk to machines, is far easier to recognize than spontaneous *conversational speech* between two humans. The quality of the signal can also be a factor in the ease of recognition: speech recorded under optimal conditions in a recording studio will of course be easier to recognize than someone speaking at a distant microphone on a busy street. Finally—and perhaps most importantly for the current work—variation in speaker classes can impact robustness of an ASR when the speaker class lies outside what the system was trained on. Speaker class includes differences in *accent* or *dialect*, as is most relevant for our purposes, but also by age and sex/gender. Creating systems robust to all of this variability is by no means a solved problem, despite the notable progress in recent decades.

Prior to the turn of the century, HMM-Gaussian Mixture Model (GMM)-based models were the de facto standard for ASR, typically consisting of four main components: Signal processing and feature extraction, acoustic model, language model, and hypothesis search (Yu and L. Deng, 2015). These systems convert of an acoustic waveform into feature vectors by a signal processer, which are then combined by a decoder with a dictionary and language model or grammar network into a recognition network. Using this network, one can calculate the most likely word sequence given the waveform of the speech input (S. M. Witt, 2000). A number of toolkits have cropped up to support development of these systems, such as Kaldi (Povey et al., 2011), and later ESPnet (Watanabe et al., 2018) as E2E architectures became more prevalent. These toolkits provided libraries and useful recipes to create reproducible systems for an array of use cases, with Kaldi in particular becoming a very popular toolkit with which to develop ASR systems. Although these are still in use, the rise of multi-purpose ML libraries like Pytorch (Paszke et al., 2019) and Hugging Face (Wolf et al., 2020) have provided a simplified alternative to training modern ASR systems.[5]

[2] https://alexa.amazon.com

[3] https://www.apple.com/siri/

[4] https://assistant.google.com/

[5] It should be noted that speech-focused libraries like SpeechBrain (Ravanelli et al., 2021) and ESPnet-EZ (Someki et al., 2024) provide a more modernized, python-focused interface similar to these general-

2.5.1 Wav2Vec 2.0

Following the rise of Transformers in ML generally, transformer-based ASR models such as *wav2vec 2.0* (commonly Wav2Vec2) have become increasingly popular for ASR tasks, including extracting phones or phonemes from raw acoustic data. Similar to other E2E alternatives, it is conceptually simpler than leading models, while its self-supervised pre-training framework achieves competitive performance with a reduced dependance on labeled data. It consists of three main components: a Convolutional Neural Network (CNN)-based *feature encoder*, a Transformer-based *context network*, and a *quantization module* (see 'Illustration of the Wav2Vec2 framework adapted from Babu et al. (2021), no alterations' on page 17). The feature encoder takes raw waveforms as input and outputs latent speech representations to be used both by the context network and the quantization module. The context network captures contextual information about the encoder output, while the quantization module divides the encoder output into discrete speech representations to be used as targets during self-supervised pre-training (Baevski, Auli, et al., 2020). During pre-training, a given proportion of feature encoder outputs (time steps) are masked, similar to the masked language model pre-training objective of Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019). For each masked time step, the model then identifies the quantized speech representation produced by the aforementioned quantization module by solving a contrastive task.

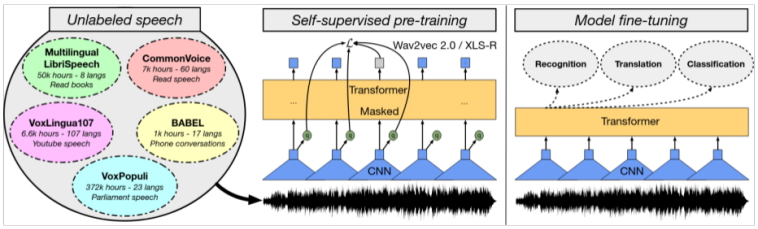


Figure 2.2: Illustration of the Wav2Vec2 framework adapted from Babu et al. (2021), no alterations

Using the above framework, Conneau et al. (2020) introduced the XLSR Wav2Vec2 model pretrained on multiple languages, with Babu et al. (2021) introducing several well known checkpoints of XLSR models pretrained on 53 languages at three different sizes of 300 million parameters, 1 billion parameters, and 2 billion parameters often short-handed as 300m, 1B, and 2B variants respectively. After pre-training on unlabeled data, these networks can be fine-tuned by adding a linear layer head of a size corresponding to the vocabulary of the task, and then optimized using Connectionist Temporal Classification (CTC) loss (Graves et al., 2006) on a much smaller set of labeled data, achieving state-of-the-art results in scenarios where labeled data is scarce Baevski, Zhou, et al. (2020). These processes can be seen in Figure 2.2 adapted from Babu et al. (2021).

ist ML libraries while retaining some popular key features from Kaldi such as recipes, which have themselves become increasingly common in generalist libraries as well.

3 Related Work

In the following section, we will dive into some existing strategies for overcoming data scarcity in MDD and ASR more generally. These approaches can be broadly split into two groups: attempts to eliminated dependency on the scarce L2 data entirely, or through generating the desired L2 data through various synthesis or augmenting techniques (Korzekwa et al., 2021). This delineation echoes the approaches to data scarcity for low-resource languages outlined in Section 2.2. Both avenues will be explored in the current work, specifically through:

1. ASR model ensembles,

–and–

2. TTS-generated data augmentation

The former will serve as our exploration into knowledge transfer to circumvent the need for scarce L2 data, where the latter instead investigates the ability to overcome this scarcity directly through data synthesis.

3.1 ASR Ensembling Strategies

As covered more extensively in the previous chapter, the state of modern ML has seen a proliferation of models trained on massive amounts of data tuned to perform well on a variety of target tasks, but which can still struggle to perform satisfactorally outside their domain of expertise. One long-standing general strategy for overcoming the limitations of any single classifier model in ASR as in ML more generally is through *ensembling*. In general terms, an ensemble refers to a set of classifiers whose outputs are combined in some way (Dietterich, 2000). Doing so effectively requires that these classifiers be similar in function but complementary, and being diverse enough to cover each other's weaknesses (Hansen and Salamon, 1990). Originally, ensembles combined outputs of constituent models through Bayesian averaging, i.e. weighing the output of the model by the posterior probability of the output, giving each model what amounts to a weighted vote in the output (Dietterich, 2000). Despite this mathematically principled origin, it proved to be prohibitively time consuming to calculate for complex problems, and so alternatives were introduced which were more computationally tractable. A wide range of ensembling techniques have emerged over the years, including boosting, bagging, stacking, majority voting and confidence voting, all of which have proven robust methods of improving performance on a variety of tasks (see l. Deng et al., 2012; Fiscus, 1997; Gitman et al., 2023; Maclin and Opitz, 1997, inter alia). For ASR, these approaches have yielded improvements over singular models with combinations of different model architectures (Arunkumar et al., 2022; L. Deng and Platt, 2014), and different training data such as for specific languages or dialects (Agrawal et al., 2023; Gitman et al., 2023, inter alia).

An early example of an equal-weight, majority voting-based system in ASR is Recognizer Output Voting Error Reduction (ROVER) (Fiscus, 1997), which aligns multiple systems with eachother to then vote on each aligned word slot, producing an aligned output string reached by model concensus. More dynamic approaches on the same

theme have been explored as with Agrawal et al. (2023), whereby a weighting module is trained on transcriptions to select expert models so as to to minimize word error rate (WER), leading to an improvement over equal-weight voting. A number of confidence-based approaches have been taken towards language identification for both HMM and more modern E2E architectures Gitman et al. (2023), whereby only the outputs of the most confident model from the ensemble are used. One issue with deriving reliable confidence measures for modern E2E deep neural network (DNN)-based models lies in their tendency toward overconfidence in ouptput predictions (Wei et al., 2022), though some mitigation strategies have been introduced to derive more reliable confidence estimates from entropy calculations (Laptev and Ginsburg, 2023). Ensembles have been utilized in a number of ways for phone-level MD as well. Ananthakrishnan et al. (2011) investigated MD with a series of binary classifiers to detect mispronunciations typical of Swedish learners of varying L1 backgrounds, focusing on systematic phoneme confusion patterns over several utterances and exeplifying through two case studies how this approach could be used to motivate targeted interventions. Calik et al. (2023) also pursued MD for Arabic phonemes using a variety of model architectures and combination techniques i.e. bagging, boosting, stacking, and majority voting, finding majority voting the most performant.

Despite the relative scarcity of ensembling applications in MD andMDD, the limitations of data and model expertise motivating ensemble applications more broadly clearly apply to L2-speech-specific phone classifiers as well: monolingual data to train expert models for Irish and English is available in abundance, but phonetically labeled data for Irish L2 speakers with an English L1 is, like phonetically-annotated L2 data more generally, comparatively scarce. Models tuned to output phones for English would not be expected to reliably transcribe Irish-specific phones, and models tuned to Irish would similarly underperform for English phones. But given the possibility of combining these expert systems in an ensemble, could each constituent expert model's phone coverage be leveraged to complement the other towards identifying mispronunciations in Irish L2 speech? If combined as a confidence-based ensemble of monolingual expert systems, we might for example expect an English model to more confidently predict English-like phones in an attempted Irish utterance with heavy interference from an English speaker's L1, while a more native-like pronunciation for other segments without any L1 interference might be confidently transcribed by the Irish ASR. If we choose the most confidently predicted phone from each span, we might be able to derive a combined string that more accurately reflects the segmental errors present in learner speech than is possible for each model on its own. Although model ensembling has a long history in ML and ASR, confidence-based ensembles have not to our knowledge been leveraged toward phone-level MDD, and although majority voting configurations have been shown to be performant alternatives, when applied to two systems with imperfect overlap in the phone inventory, majority voting can simply devolve into ties with no obvious resolution. We opt instead to select the model most confident in its prediction per span, letting the expert models select among the phones it is trained on.

3.2 Data Generation with Synthetic Speech

On the other end of common solutions to L2 data scarcity, we encounter *data augmentation* and *data generation*. Data augmentation is frequently done through labeling of existing audio (see Bartelds et al., 2023; Xu et al., 2020; Yang et al., 2022, inter alia) or altering existing audio through targeted alternations to phonemes or other perturbations (Kheir et al., 2023). Automated labeling of unlabeled audio presents a promising

avenue for low-resource languages, as unlabeled audio is easier to collect than manually annotated data (Bartelds et al., 2023), and for languages with at least some digital presence, there may be latent reservoirs of this kind of data available to propel such an approach. In ASR generally, Hayashi et al. (2018) pursued back-translation-like data augmentation to bolster performance, whereby a text-to-encoder translation model was trained to predict ASR hidden states, which were then used to retrain the decoder without need for any speech paired to the augmenting text data. For MDD, Yang et al. (2022) illuminated the use of momentum pseudo-labeling via a teacher-student model ensemble using Wav2Vec2 as a promising way to leverage plentiful unlabeled L2 speech for improving MDD performance.

In recent years, however, as end-to-end ASR models like Wav2Vec2 become increasingly capable, so to do their TTS counterparts, advancing to the point where the speech they produce sounds nearly as natural as human speech. This development has led a number of researchers to pursue synthetic data generation via TTS as another way to fill the data gap for low-resource ASR (see Bartelds et al., 2023; Fazel et al., 2021; Thai et al., 2019, inter alia). Use of TTS outputs to augment data has been demonstrated to provide improvements in WER for ASR models (Bartelds et al., 2023), with the caveats that not all languages have readily available TTS systems. For MDD, however, this inequality in TTS availability is less of an obstacle to providing useful data. The task of generating mispronunciation data has been framed in recent years by Korzekwa et al. (2022) and Zhang et al. (2022) as *phoneme paraphrasing*, as an analogue to the paraphrasing task in Natural Language Generation (NLG). The former approach by Korzekwa et al. (2022) proposes three avenues for synthesis—phoneme-to-phoneme (P2P), TTS, and sequence-to-sequence (Seq2Seq)—all improving accuracy in detecting mispronunciation errors. The latter approach focuses on a P2P pipeline, where canonical phone strings are translated to L2-like variants using a Seq2Seq MT model, which is then diversified using their diversified and preference-aware decoding (DPAD) module to improve generalizability, and finally fed to a TTS system L1 to generate L2-like mispronounced audio. Critically, a seed set for training the Seq2Seq paraphraser was extracted from the expertly-annotated dataset used L2-arctic, resources not necessarily available in our case for Irish. Similarly, the preference term enriching the variety of their DPAD system relies on curated sets of mispronunciation patterns specific to L1 Mandarin speakers.

Similar to Zhang et al. (2022) we see the potential for synthetic data as a possible answer to MD data scarcity through Seq2Seq translation to create controllable errors mirroring what one would expect from language learners with the same L1 the TTS is tailored to. Mapping phonetic transcriptions of canonical Ulster pronunciations to inputs acceptable to an English TTS gives highly English-sounding approximations of Irish speech with exactly the kinds of segmental errors one expects from English L1 learners of Irish.[1] Instead of using a NN-based architecture for our phone-to-phone paraphrasing, we use a monolingual English ASR model fed with Irish speech for which we have corresponding phonetic transcriptions. As we do not have an expertly annotated seed corpus to draw from with Irish as their work does by drawing the seed data from L2-arctic, we considered a zero-shot approach that circumvents the need for expensively annotated L2 data, an approach which we will elaborate on in the following chapter. L2-gen provides greater diversity through their DPAD augmentation system, but in the interest of time, we do not attempt here to introduce additional variance to the TTS voices. However, we recognize this innovation as a critically im-

[1] Another noteworthy property of the recordings is the English-like prosody and stress patterns, which could be pursued in an investigation into suprasegmental deviation from Ulster speech, but lies outside the scope of the current work.

portant one for promoting more robust performance and will explore possible avenues for mimicking this behavior with an ASR-based system in Chapter 8.

4 Methods & Materials

In the following chapter, we will elaborate the core experiments explored for the current work: one which explores ensembling monolingual models as described above as a solution to MDD in the face of data scarcity; the other explores data augmentation via an English ASR model transcribing TTS inputs from Irish speech to create the mispronounced data required to train an ASR model directly for MDD. We will begin with detailing the features in common between these experiments before continuing to each experiment in turn, and end with the evaluation details shared between experiments.

4.1 General Experiment Setup

Both of the MD pipelines explored follow the same general flow, starting with audio input of L2 speakers speaking Irish. The input waveform is then processed by one or several ASR models configured to output strings of phones. These phone strings are then aligned to canonical IPA transcriptions of the target string taken from a phonetic grapheme-to-phoneme (G2P) model, from which we can derive a phonetic string from the graphemes in the orthographic form of a word, with corresponding audio scraped from the online Irish pronunciation dictionary Teanglann, to be fully described in Section 4.2.1. Manually annotated IPA strings serve as our ground truth labels, which are also aligned to the canonical string, and then scored for mispronunciation, a procedure we will expand upon in Section 4.4.

The model chosen for all experiments was the 300 million parameter version of Wav2vec2 XLS-R (Babu et al., 2021). This choice of architecture was motivated by the desire to leverage the aforementioned pre-training/fine-tuning paradigm on a model shown by Babu et al. (2021) to improve on previous work most notably for low- and mid-resource languages. Selection of the 300 million parameter version of this architecture was primarily motivated by desire to rapidly train models and test ideas, aiming for proof-of-concept over beating the state-of-the-art. These models were fine-tuned on phonetically annotated datasets, allowing us to obtain phone-level transcriptions for comparison.

All models were implemented using the Hugging Face *Transformers* library (Wolf et al., 2020). Training was managed with the Transformer *Trainer* class, configured via *TrainingArguments*. Training progress and metrics were logged to Weights & Biases (Biewald, 2020). Our implementation of Wav2Vec2 for phonetic output followed the publicly available tutorial by Vitouphy on Kaggle.[1] Models were instantiated with an attention dropout of 0.1, layer dropout of 0, feature projection dropout of 0, mask time probability of 0.75, mask time length of 10, mask feature probability of 0.25, and mask feature length of 64 adhering to the default values suggested by the tutorial.

Training for all ASR models used identical parameters, with batch size of eight, learning rate 3e-5, 2000 warmup steps, and 16-bit precision training to reduce memory usage and speed up training. Minimizing character error rate (CER) was used as the evaluation metric and in guiding early stopping, which had a patience of 6 epochs

[1] https://www.kaggle.com/code/vitouphy/phoneme-recognition-with-wav2vec2

with a maximum of 50 epochs of training to avoid overfitting. Training was carried out on kaggle[2] using their P100 GPU. Full configuration details are given in Table 4.1.

Table 4.1: Model architecture and training configuration shared across all experiments.

Category	Parameter	Value
Architecture	Base model	XLS-R 300M
	Attention dropout	0.1
	Layer dropout	0.0
	Feature projection dropout	0.0
	Mask time probability	0.75
	Mask time length	10
	Mask feature probability	0.25
	Mask feature length	64
	CTC loss reduction	mean
Training	Batch size	8
	Gradient Accumulation Steps	4
	Learning rate	3e-5
	Evaluation strategy	Epoch
	Learning Rate warmup steps	2,000
	Learning Rate scheduler	linear
	Precision	16-bit
	Objective	CER
	Early stopping patience	6 epochs
	Max epochs	50
	Optimizer	Adam

4.2 Experiment 1: Monolingual Model Ensembling

In this section we will begin by describing the data used, first for the English model, then the Irish model. Following this we touch briefly on the preprocessing procedure before expanding on the full experimental setup.

4.2.1 Data

To train our monolingual systems, we procured two monolingual corpora of read speech audio with phonetic transcriptions. Our English model was trained on the TIMIT corpus Garofolo et al. (1993), as it provides valuable phonetic transcriptions for American English speech. The Irish model trained on audio data from the online Irish Dictionary and Language Library[3] with derived phonetic transcriptions which will be detailed below.

TIMIT

The TIMIT corpus is a read speech corpus of American English speakers who spent their childhoods in one of eight major dialect areas of the United States. These areas are widely recognized with the exception of the Western dialect region, where boundaries are not confidently delineated, and the "army brat" group, consisting of speakers which frequently moved during their childhood due to the demands of highly mobile

[2] https://www.kaggle.com
[3] https://www.teanglann.ie/en/

military service member parents, resulting in exposure to a variety of dialects (Garofolo et al., 1993). The full dataset consists of 6300 sentences from 630 speakers, so ten sentences per speaker. The duration of the data loaded from the Kaggle dataset linked to the TIMIT github dataset[4] lies at two hours and forty minutes. Note that this is not the full dataset, but to ensure some level of comparability between the monolingual models so as to not advantage one over the other, this data was left deliberately downsampled. The TIMIT phonetic transcriptions were given in a revised version of ARPABET, a set of transcription codes developed by Advanced Research Projects Agency (ARPA) (for the original set, see Rice, 1976).

Teanglann

The Dictionary and Language Library is an online resource developed by Foras na Gaeilge[5] in conjunction with the New English-Irish Dictionary. Among the resources available are The Pronunciation Database, which contains recordings of individual words spoken by native speakers from three major dialects: Connacht, Ulster, and Munster. We elected to focus on one dialect, Ulster, as the target pronunciation for our MDD task, but we of course recognize the need for such tools for other dialects as well. The Ulster recordings used for our monolingual Irish model were acquired via webscraping adhering to the site's robots.txt file limit of one request per two seconds. The words scraped were drawn from an available G2P dictionary to map the orthographical forms to to ensure scraped audio had corresponding canonical IPA transcriptions.[6] The resultant combined dataset had an audio duration of roughly five hours of audio.

Preprocessing

The audio from both Teanglann and TIMIT was loaded and resampled to 16kHz for compatability with Wav2vec2. For the phonetic trancriptions, TIMITs phonetic representations were translated to IPA using the phonecodes library, and maintaining one space between each phone or dipthongs.[7] The Irish data was transcribed through a lookup in the aforementioned G2P dictionary, which adhered to the same spacing rules as for TIMIT. For the Irish set, phones from the g2p canonical forms had all but palatal diacritics collapsed to the base phone to minimize the label space while preserving the particularly salient palatalized/velarized distinction from Irish phonology (see Section 2.1). We will return to this choice in Chapter 6.

We split the datasets into train/dev/test splits with .8/.1/.1 ratios respectively, which were then trained according to the procedure described in Section 4.1. One critical limitation with the Irish data was that due to the acquisition method, we were left with no way verifiable way to identify speakers, so the different splits likely contains speaker overlap. We sought to mirror this limitation with the English data so as to not bias any particular language when ensembling them, and so the monolingual evaluation metrics are also likely overly optimistic.

[4] https://github.com/philipperemy/timit

[5] Foras na Gaeilge is a group which promotes the Irish language, supports Irish-medium education, and advises public and private sector organizations, among its other functions. https://www.forasnagaeilge.ie/

[6] The dictionary of words to scrape was derived from an Ulster Irish G2P file generously provided by Jim O' Regan

[7] dipthongs were kept together as a single single unit to keep with standard practice

4.2.2 Experimental Setup

The two core models in the ensemble were Irish and English monolingual ensembles trained on the datasets described above. To combine their outputs, we need to first establish that we are comparing confidence values across the same stretch of audio. To do this, we first derived canonical spans from the torch audio forced alignment library[8] using the Irish model and the canonical transcriptions of the input, since that was the only model available to us trained on the canonical phone inventory, which is needed for a model to assign any span to members of that inventory. Since the output spans using this function on subword units spans only a few frames, we used these output frames as the middle point to expand the canonical spans until they met a neighboring span as a heuristic approach, though there is room for experimentation. At this point we found each model's most confident phone over each span along with its corresponding confidence value, and compared the values to determine which model's output we would use for each span's slot in the final decoded string. The baseline confidence approximation used was raw softmax probability, but as these probabilities are prone to overconfidence, we explored confidence calculated from Gibbs entropy following Laptev and Ginsburg (2023) and Gitman et al. (2023) as seen in eq. (4.1) where H_g is the Gibbs entropy, V is the vocabulary size for our model, p_v is the probability the model assigns to phone v in a given frame, and $F_g(p)$ is the Gibbs entropy confidence. Aside from raw confidence comparison we tested the use of an Logistic Regression (LR) model selection block following Gitman et al. (2023) to mediate the confidence values in case one model was systematically more confident than the other. This LR block was trained on 99 manually annotated word-level utterances held out from the evaluation data described in more detail at the end of the chapter.

$$\mathrm{F}_g(p) = 1 - \frac{\mathrm{H}_g(p)}{\max \mathrm{H}_g(p)} = 1 + \frac{1}{\ln V} \sum_{v=1}^{V} p_v \ln p_v \tag{4.1}$$

Due to the relatively granular phonetic distinctions separating some Irish phonemes, particularly palatalized consonants, we expected confidence values to be systematically lower for the Irish model for these palatalized/velarized pairs than English would in discriminating the class as a whole. One solution explored to overcome this issue was to add a Russian model to the ensemble as a kind of high-resource palatalization authority, since the language shares a considerable number of palatalized phonemes with Irish. In the ensemble procedure described above, the Russian model outputs, when included, were chosen automatically if the output phone was palatalized. This model[9] was a Wav2Vec2-large-xlsr-53 variant taken from the Huggingface Model Hub and was fine-tuned on the commonvoice Russian dataset to output phonemes, which could be readily mapped to IPA by mapping dictionary included with the model.

Another potential solution to the problem of diluted model confidence was explored in simply pooling the values of related phones (e.g. [l] & [lʲ]) when choosing between the models, and then the most probable phone among those if the Irish model was chosen. The intuition guiding this can be visualized below4.1.

4.3 Experiment 2: Synthetic Mispronunciation Data

In this section we will confine ourselves to describing the how the synthetic data was derived and processed into a training set. The model itself was trained on this

[8] torchaudio forced_align documentation
[9] HuggingFace model card

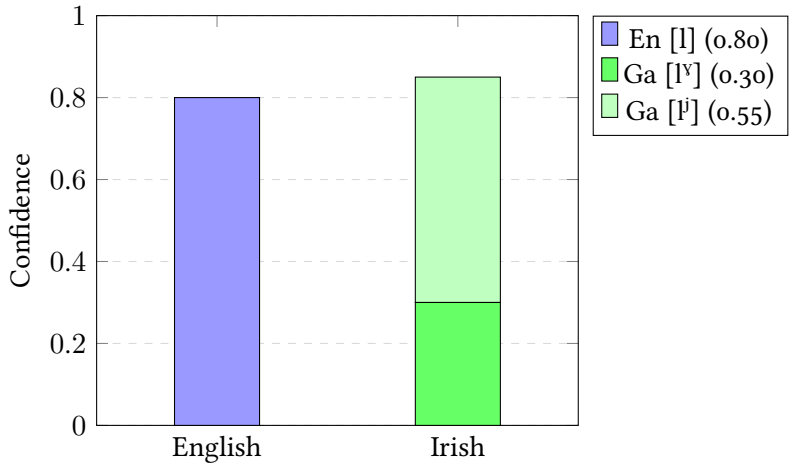


Figure 4.1: Phone pooling for a span with canonical [lʲ]. Without pooling, the Irish model's split distribution would give it lower confidence than English and lose. Summing probability mass across the broad/slender family (shaded, 0.85 total) beats English (0.80), and the Irish model's top prediction [lʲ] is selected.

set identically to the procedure described at the beginning of the chapter. Decoding functions identically to the ensemble, whereby the Irish monolingual model is used to force align the canonical string, defining the spans from which we search for the most confident output phone.

4.3.1 Data

Canonical recordings were obtained from the Teanglann pronunciation dictionary, with corresponding canonical IPA transcriptions from the aforementioned Ulster Irish G2P dictionary: the same dataset as that used for training the Irish model in Experiment4.2. This canonical data was complemented with augmenting learner-like recordings obtained via Google Cloud text-to-speech (Google Cloud, 2026) using the full set of American wavnet (van den Oord et al., 2016) voices at a recording frequency of 16kHz. These were chosen due to the accepted characters for Speech Synthesis Markup Language (SSML) input being most similar to our monolingual English ASR outputs: a model trained on American English speaker recordings. Unlike the Seq2Seq phone paraphrasing model used by Zhang et al. (2022), we opted for a simple ASR approach to acquiring SSML inputs. To do this, we fed our canonical Irish recordings through the monolingual English model described above, whereby we obtained transcriptions of realistic English-L1 mispronunciations of the Irish speech. The input was given in SSML to enable direct use of IPA transcriptions to guide pronunciation.

These mispronounced synthetic samples together with the canonical Irish recordings of each word composed the *positive* and *negative* utterance pairs of the training data, similar to the *adversarial sampling* strategy used by Zhang et al. (2022) shown to increase model sensitivity to MD by grouping mispronounced and correctly pronounced samples during training. In contrast to their approach, however, we include only one adversarial utterance in the training sample. The transcriptions and recordings of each pair were concatenated with ordering of the source randomized.

4.4 Evaluation

In the following section we detail the evaluation data used for evaluating each experiment, as well as the process used to annotate it. Following this, we describe our

evaluation procedure and the metrics and statistical test used.

4.4.1 Data

To measure an MDD system's sensitivity to plausible Irish L2 errors, we first needed a source of Irish L2 speech. We began with the Irish portion of the Few-shot Learning Evaluation of Universal Representations of Speech (FLEURS) dataset (Conneau et al., 2022). The crowdsourced nature of the dataset allowed for speakers of various skill levels to contribute, an attribute that is perhaps not attractive for most ASR applications, but which perfectly suits our purpose, as there were some markedly non-native recordings in the evaluation set. Canonical transcriptions of the data was derived from the aforementioned G2P dictionary, with a fallback lookup for out of vocabulary (OOV) consisting of an Montreal Forced Aligner (MFA) G2P model also trained on this dictionary. The minority of sentences left with unknown tokens were removed, which largely stemmed from proper nouns consisting of letters outside of what the G2P fallback could handle. Audio volume was normalized with ffmpeg (FFmpeg Team, 2024), specifically ffmpeg-normalize at default settings. White-noise handling was carried out with ffmpeg Adaptive Fast Fourier Transform Denoise with a noise floor of -50 dB, white noise for noise type, and otherwise default settings.

To prepare the FLEURS dataset for use in phone-level MDD, a small subset of the audio data was manually segmented and transcribed in IPA to establish a ground truth set of transcriptions using a Label Studio html interface created by the author (an example of which can be seen below in Figure 4.2) to transfer data to different machines via Google Cloud Service (GCS). Of the audio segments produced in this way, 99 were used as training data for the aforementioned LR selector block in Section 4.3, and the remaining 146 were used for evaluating both experiments. The process of capturing these deviations consisted of the following steps: the audio was loaded into an interface with the canonical IPA transcriptions as well as orthographic representations included in the interface; recordings were then repeatedly assessed for natural breaks in speech that could be used to anchor clear segments to; these segments were then repeatedly analyzed for perceived realization within the bounds of valid English or Irish phones used for model training, referring to an interactive IPA chart with sounds of each phone[10] when unsure about the sounds produced. This process was carried out by the author only, a non-native Irish speaker of modest ability, with no professional phonetician training, and with limited reference material to refine my annotations on with respect to palatalized/velarized consonants, all notable limitations we will return to later. The choice to have the target canonical string visible during annotation was not free of trade-offs: negative effects on the annotation process will also be discussed in chapter Chapter 6 .

4.4.2 Evaluation procedure

We evaluate phone-level discrimination of the ASR by first aligning canonical strings to the ground truth annotated strings with the Needleman-Wunsch algorithm (Needleman and Wunsch, 1970) with the PanPhon (Mortensen et al., 2016) library in place of uniform mismatch costs to help better align sequences with articulatorily similar phones. Alignment of the model outputs and canonical strings to the ground truth string are the same length as a consequence of the canonical string defining model decoding spans for each experiment through False Acceptance (FA). This alignment by architecture then trivializes the alignment of model outputs to corresponding ground truth strings, as they are equal in length to the alignment of the canonical string to

[10] ipachart.com



Figure 4.2: Annotation interface created in Label Studio to assist in phonetic annotations. Annotator manually defines segments, drags the canonical representation of the segment to the canonical slot, and annotates perceived production in the ground truth slot.

the ground truth string.[11] This same feature renders the whole decoding system architecturally blind to insertions however, as models can not decode a phone outside the force aligned spans—an issue we will discuss in Chapter 6.

4.4.3 Metrics

For *detection*, we can focus on alignment between canonical and ground truth to True Acceptance (TA) True Rejection (TR), False Rejection (FR) and FA. At each canonical phone position, the correctly pronounced/mispronounced verdict is given by the canonical/ground truth alignment. Then, given the canonical/model hypothesis alignment, TA refers to those also determined to be correct by the model, while FR refers to those which are deemed incorrect by the model but correct by the canonical/ground truth. Among the pronunciations deemed *mispronounced* by the canonical/ground truth alignment, those accepted by the model/canonical alignment are considered correctly pronounced as FAs, while those considered mispronounced are accurately rejected as TR. At the level of diagnosis, we are interested in further analysis of those pronunciations which fall under TR, specifically for the purpose of segmental pronunciation analysis: for each TR, is each phone detected by the ASR model in agreement with the aligned ground truth string? Those that are get labeled with Correct Diagnostic (CD), and those which are not, Diagnostic Error (DE). An illustration of this evaluation hierarchy is available in Section 4.4.3.

Similar to prior MDD research (Leung et al., 2019; Li et al., 2017; Qian et al., 2010, inter alia), we focus on F1-score, False Rejection Rate (FRR), False Acceptance Rate (FAR), and Diagnostic Error Rate (DER) to evaluate performance. Precision, Recall, and F1-score are calculated in Equations (4.2a) to (4.2c) and False Rejection Rate (FRR), False Acceptance Rate (FAR), and Diagnostic Error Rate (DER) are calculated from the values above in Equations (4.2d) to (4.2f). Precision and Recall are not included in the presentation of results in the interest of focusing on the metrics more directly meaningful to MDD, overall error rate, given as the sum of FAs, FRs, and DEs divided by the total number of predictions in Equation (4.2g). This corresponds to the red leaves in Section 4.4.3, with correct predictions corresponding to the green leaves.

[11] It does matter which string you align the ground truth to first for pairwise alignment like this.

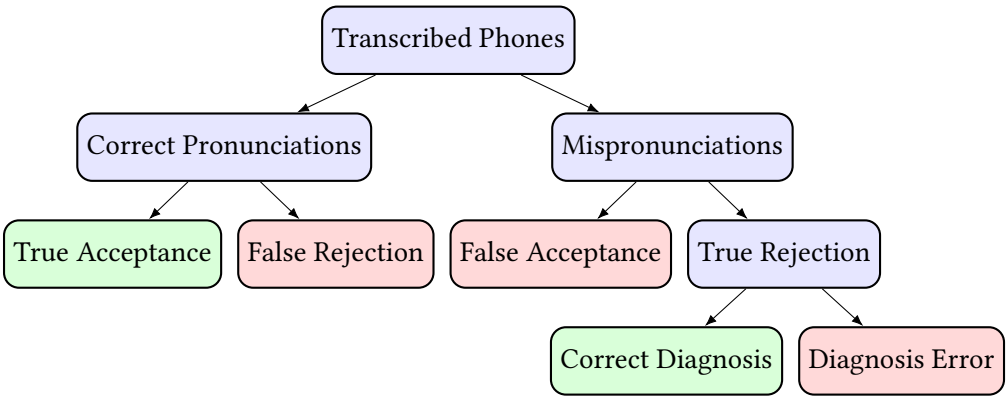


Figure 4.3: General Evaluation hierarchy of phone level MDD. The lowest level on the right is where diagnosis happens for correctly rejected phones: were the phones detected and rejected by the system actually those present in the mispronounced ground truth labels?

$$P = \frac{TR}{TR + FR} \quad (4.2a) \qquad R = \frac{TR}{TR + FA} \quad (4.2b) \qquad F1 = \frac{2 \cdot P \cdot R}{P + R} \quad (4.2c)$$

$$FRR = \frac{FR}{TA + FR} \quad (4.2d) \qquad FAR = \frac{FA}{FA + TR} \quad (4.2e) \qquad DER = \frac{DE}{CD + DE} \quad (4.2f)$$

$$Err = \frac{FA + FR + DE}{N} \quad (4.2g)$$

For statistical testing, we utilized McNemar's test for classifier comparisons from the mlxtend library (Raschka, 2018) on the overall error rate detailed above. This test can be used for machine learning to compare predictive accuracy of two classifiers (Dietterich, 1998) which suits our comparisons of various systems against a monolingual baseline. For this, we evaluate each model on a test set and count the instances where they correctly or incorrectly detect or diagnose mispronunciations and construct a contingency table with 4 cells corresponding to each combination of model classifications, the relevant values being b and c, or the off-diagonal disagreement counts between models compared.Table 4.2

	Baseline correct	**Baseline incorrect**
Model correct	a	b
Model incorrect	c	d

Table 4.2: McNemar contingency table for paired significance testing of two systems evaluated on the same set of phone positions. Each cell counts positions where the respective combination of outcomes occurred. Only the discordant cells matter for the test: b (system correct, baseline wrong) and c (system wrong, baseline correct).

For the null hypothesis to hold, neither model performs better than the other, which formally translates to the probabilities p(b) and p(c) from Table 4.2 being the same. The test statistic can be calculated as seen in Equation (4.3) and can, after setting a significance threshold $\alpha = 0.05$, be used to calculate a p-value, or the probability of observing the counts b and c under the null hypothesis $p(b) = p(c)$. If the p-value lies under the significance threshold, we can reject the null hypothesis.

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \tag{4.3}$$

5 Results

In this chapter we will present the results of the the above experiments. The evaluation performance of each model on its evaluation dataset in terms of CER is given in table 5.1. We then continue with the synthetic experiment results, then on to the ensemble experiment results, before presenting the most performant model from each experiment alongside the baselines. Finally, we will present each ablation's performance with respect to the full version of each experiment in terms of overall difference in error rate.

Table 5.1: Constituent model evaluation results

system	Eval CER
Irish monolingual model	0.167
English monolingual model	0.124
Irish+English multilingual model	0.161
L2 model unpaired	0.112
L2 model paired	0.089

For the synthetic experiment results given in Table 5.2, one can see the most performant model by way of overall error rate was the unpaired ablation, though the model trained with contrastive paired samples did better by way of FAR and DER. Despite this, given our $\alpha = 0.05$ significance cut-off, no model outperformed baseline to a statistically significant degree.

For the Ensemble experiment results given in both Table 5.4 and Table 5.3 for those with and without a selector block respectively, the most performant model by way of F1 score is the multilingual Irish+English baseline model trained on a combined set of the two languages. The 2-way ensemble with a probability confidence function was the most performant among models without a selector block as seen in Table 5.3. However, even the least performant of the selector block models outperformed it. The most performant selector block model was the 2-way model with the Gibbs confidence function and pooled confidence values. All models underperform baseline to a statistically significant degree as seen in Table 5.4.

For the main results, one can see how the Irish baseline compared against the multilingual model, the unpaired synthetic model, the paired synthetic model, and the most performant ensembles with and without a selector block in Table 5.5.

For relative ablation performance, we can see for the synthetic experiment that the removal of adversarial sampling improved the error rate by by 0.036 compared to the

Table 5.2: Mispronunciation detection results: synthesis experiment. * indicates p < 0.05, † indicates higher error than baseline.

System	F1	FAR	FRR	DER	Err.	McNemar
Irish only baseline	0.559	0.244	0.342	0.687	0.454	—
L2 model unpaired	**0.580**	0.240	**0.310**	0.734	**0.440**	**0.265**
L2 model paired	0.562	**0.191**	0.386	**0.660**	0.476	0.188†

Table 5.3: Mispronunciation detection results: ensemble ablations (no selector). * indicates $p < 0.05$, † indicates higher error than baseline.

System	F1	FAR	FRR	DER	Err.	McNemar
Irish only baseline	0.559	0.244	0.342	0.687	0.454	—
Irish+English	0.552	0.206	0.392	0.678	0.485	0.045*†
2-way, Gibbs	0.510	0.141	0.546	0.662	0.590	0.000*†
2-way, Gibbs+pool	0.512	0.156	**0.526**	0.719	0.589	0.000*†
2-way, prob	**0.515**	**0.130**	0.545	**0.654**	**0.586**	0.000*†
2-way, prob+pool	0.511	0.153	0.531	0.712	0.591	0.000*†
3-way, Gibbs	0.509	0.141	0.548	0.662	0.591	0.000*†
3-way, Gibbs+pool	0.511	0.156	0.527	0.719	0.590	0.000*†
3-way, prob	**0.515**	**0.130**	0.546	**0.654**	0.587	0.000*†
3-way, prob+pool	0.510	0.153	0.532	0.712	0.592	0.000*†

Table 5.4: Mispronunciation detection results: ensemble ablations (with selector). * indicates $p < 0.05$, † indicates higher error than baseline.

System	F1	FAR	FRR	DER	Err.	McNemar
Irish only baseline	0.559	0.244	0.342	0.687	0.454	—
Irish+English	0.552	0.206	0.392	0.678	0.485	0.045*†
2-way, sel/prob	0.518	**0.168**	0.499	**0.638**	0.552	0.000*†
2-way, sel/prob+pool	0.520	0.176	0.487	0.676	0.552	0.000*†
2-way, sel/Gibbs	0.519	0.172	0.494	0.641	0.549	0.000*†
2-way, sel/Gibbs+pool	**0.525**	0.179	**0.472**	0.693	**0.545**	0.000*†
3-way, sel/prob	0.517	**0.168**	0.501	**0.638**	0.553	0.000*†
3-way, sel/prob+pool	0.519	0.176	0.488	0.676	0.553	0.000*†
3-way, sel/Gibbs	0.518	0.172	0.495	0.641	0.550	0.000*†
3-way, sel/Gibbs+pool	0.524	0.179	0.473	0.693	0.546	0.000*†

full model in Table 5.6. For the ensemble, the removal of the selector block reduced error rate the most by 0.045. Removal of pooling reduced error rate by 0.004, replacing Gibbs confidence with raw probability reduced error rate by 0.007, and removal of the Russian model instead improved error rate by 0.001 as seen in Table 5.7.

Table 5.5: Mispronunciation detection results: main comparison. * indicates p < 0.05, † indicates higher error than baseline.

System	F1	FAR	FRR	DER	Err.	McNemar
Irish only baseline	0.559	0.244	0.342	0.687	0.454	—
L2 model unpaired	**0.580**	0.240	**0.310**	0.734	**0.440**	**0.265**
L2 model paired	0.562	0.191	0.386	0.660	0.476	0.188†
Irish+English	0.552	0.206	0.392	0.678	0.485	0.045*†
2-way, prob	0.515	**0.130**	0.545	**0.654**	0.586	0.000*†
2-way, sel/Gibbs+pool	0.525	0.179	0.472	0.693	0.545	0.000*†

Table 5.6: Ablation study: synthesis experiment.

System	Err.	Δ
Full (L2 model (paired))	0.476	—
– adversarial sampling	0.440	-0.036

Table 5.7: Ablation study: ensemble experiment.

System	Err.	Δ
Full (Ensemble (3-way, sel/gibbs+pooled))	0.546	—
– selector	0.590	+0.044
– Russian	0.545	-0.001
– pooling	0.550	+0.004
– gibbs	0.553	+0.007

6 Discussion

As shown above, the synthetic-augmented models did not produce results that differed significantly from the monolingual baseline, and the ensembles plus multilingual model significantly underperformed the monolingual baseline. In this chapter, we will begin by exploring differences in relative performance between ablations within each experiment to examine componental contributions to performance. Following this, we will explore confounds that could have contributed to overall underperformance of the present experiments, which we will consider from implementational to structural in nature.

6.1 Experiment 1: Model Ensembling

The ensemble results show some relative behavior between systems worth digging into. The selector block contributed the most to full system performance, but it is noteworthy that using Gibbs over raw output probability resulted in *decreased* F1 for the ensembles without a selector block, while *increasing* F1 for those with a selector block, especially when paired with pooling. This seems to correlate with the relative contributions of the English and Irish models to the ensemble with different confidence calculations, with or without a selector, and with pooling added to the combination as seen in Figure 6.1. The same components which increase performance also seem to balance the selection between models, in particular the selector block. That the selector block serves to balance the model selection comes as no surprise given that was its intended function, but it is worth reiterating its contribution to performance in this context.

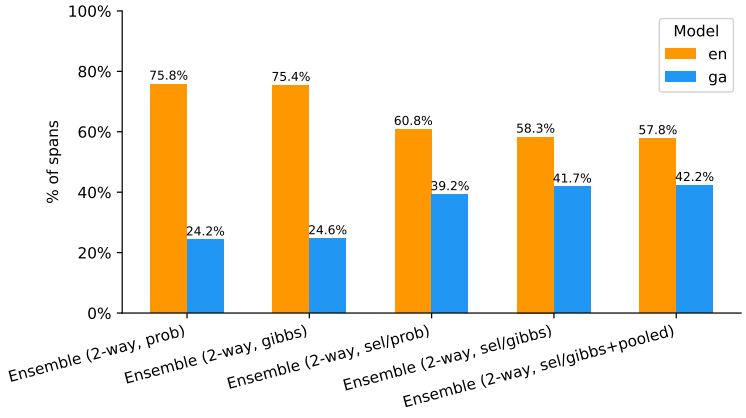


Figure 6.1: Ensemble winner contributions, en/ga

Removing the russian model improved performance slighly, but it is worth remembering that the russian role in the ensemble was as a palatalization specialist, and in the most aggressively implemented variant possible: if the russian model detected a palatalized phone, that was what got added to the final string without voting between the other ensemble members. As a more equal member of the ensemble that participates in voting, perhaps by combining its palatalization evaluation with Irish or

contributing towards majority voting, it might contribute to performance rather than it hinder it. However, if one compares the palatalizations detected by the russian inclusion in its most aggressive form against those detected by the pooling strategy, one sees that they are solving the same problem about 82.4% of the time in Figure 6.2, with russian inclusion detecting slightly more than the pooling strategy when alone, but also contributing with some false positives as we can see due to its relatively lower performance. This suggests pooling might be a lighter weight alternative to a specialist model in boosting palatalization detection in an ensemble configuration. Since every model addition increases memory use and inference time, compared to a relatively simple mapping table with pooling, this trade-off might be worth considering closely.

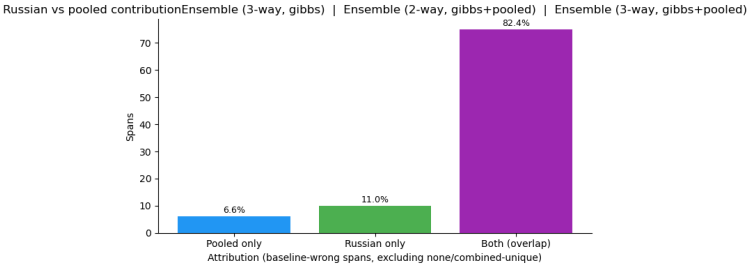


Figure 6.2: Relative performance in palatalization detection between Russian inclusion, pooling, and both in percentage terms.

Although ensemble components improved on the simplest version of the ensemble, the multilingual model and monolingual baseline still outperformed all ensemble versions. This suggests that a multilingual model might be a better starting point for combining monolingual data than an ensemble of monolingual models for future exploration.

6.2 Experiment 2: Synthetic Mispronunciation Data

Beginning with the synthetic ablation, we can see that the removal of adversarial sampling (i.e. concatenation of canonical and synthetic pairs of speech and text) improved performance relative to the full model. Training instances for this ablation were thus single word utterances, which matched the evaluation data segments also consisting primarily of single word utterances. Although this component was shown to contribute the least to the out-of-domain experiment by Zhang et al. (2022), it is worth considering that the mismatch between training and evaluation format for the full model may have degraded performance relative to the ablation in our experiment despite outperforming it when evaluating the models on data derived from the same source as their respective training data. Given these limitations, due caution should be exercised in discounting it's potential contribution in other settings.

Another possible explanation for system underperformance could lie in the absence of a TTS diversification mechanism similar to DPAD in Zhang et al. (2022). Intuitively, only capturing one mispronunciation variant does not give a full sense of variation for mispronunciations of a word. Augmenting our model data with a fuller range of possible mispronunciations for each word should bolster performance and we will discuss a possible adaption of this concept to our ASR-based approach in Chapter 8.

6.3 Implementational Confounds

Whatever might account for relative differences between models, their uniform underperformance with respect to baseline calls for a closer examination of possible causes common to both experiments. Firstly, our decision to decode strings based on canonical-anchored Force Alignment spans for all models was not a decision without trade-offs. Pinning spans to the baseline Irish model advantages the baseline since all decoding is performed based on where the baseline decides the spans are. This imperfect decoding strategy was initially intended to address the ensemble problem where aligning free-phone decoded strings with basic mismatch distance led to implausible string alignments that spuriously aligned the tails of one string with the head of another, and heavily punished alignment of phonetically close characters differing only by a few features. This was especially problematic for characters of particular interest like palatalized consonants, which would not reliably match with each other.

One way to solve this problem, and the one we chose here, is to define spans across the audio via Forced Alignment and compare ASR outputs within each span. Aligning multiple ASR model spans was itself not straight forward. CTC trained models exhibit 'peakiness' where spans of individual characters are typically confined to one or two frames (See Zeyer et al., 2021, for a fuller description). Where these peaks occur has little consistency between models. In some cases, they would come at the onset of a sound, other times the end, and some times somewhere in between. Determining reliable phone spans from these frames in CTC models is a well known problem, and our chosen heuristic was to define the canonical frames as middle anchors from where the span was expanded outward until it encountered a neighboring span or the end of the string. Our strategy produced plausible spans and output strings, but also renders the system blind to insertions; if the speaker inserts sounds between where the canonically anchored spans expect only one, the system chooses the most confident candidate and discards the rest. This shortcoming also illuminated a particular shortcoming in the aforementioned decision to collapse the Irish phone set to phones stripped of all diacritics except palatalization markers. Some word-initial velar consonants were annotated as simple velar consonants in the Irish G2P model, but are realized with a velar offglide that the ensemble and synthetic models could detect and labelled as a separate phone. This was likely due to the English model not collapsing the velar offglide together with the preceding phone as the Irish model was trained to do. One can clearly see the implications in this point of failure combined with insertion blindness in Figure 6.3. The limitations above point to a number of failure points worth

Ensemble (2-way, prob) | 10266820967862372597_l5ruv

canonical	f	i	nʲ	uː	—
gold	f	i	nʲ	uː	vʲ
asr	f	w	n	vʲ	—

Figure 6.3: Ensemble output where velar offglide /w/ edges out /i/ and /vʲ/ edges out /uː/ due to canonical anchoring and insertion blindness.

addressing. If we are to stick with canonically-anchored Forced Alignment spans for our evaluation, determining spans heuristically from CTC model peaks is likely not the best approach. It would likely be beneficial to use MFA models to provide more reliable phone spans to be mapped onto the frames of the CTC model from which we can then extract phones. Insertion blindness is not a necessary consequence of Forced

Alignment methods, rather an implementational compromise that simplified aligning the ASR outputs to the canonical string. One way around this might be to do a version of free-phone decoding where we align the string outputs in a span to the canonical string, but at that point, we are most of the way to free-phone decoding and string alignment. Pursuing this might be worth considering if using feature distances from libraries like PanPhon in the place of uniform mismatch penalties for alignment, as this is a much conceptually simpler approach that handles granular distinctions like Irish palatal/velar contrasts with minimized mismatch penalties: precisely the problem that motivated Forced Alignment span anchoring in the first place. We already align the ground-truth string to the canonical in this way, and switching the ensemble alignment to this strategy would bring it conceptually closer to classic ROVER configurations whereby one progressively aligns constituent models to eachother into a Word Transition Network. One final limitation worth reflecting on is how to handle tokenizer mismatch such as between Irish encoding velar offglides as one token vs English treating them as two. This could be solved with a simple mapping of velar or palatal offglides to the nearest Irish equivalent at evaluation, but carefully considering annotation and corpus reliability could avoid the need for such post-hoc stopgaps alltogether.

6.4 Structural Limitations

6.4.1 Annotation Reliability

Though efforts were made in constructing the annotation interface and set up to ensure consistent annotations, lack of phonetician training undercut the quality of the evaluation set and its comparability with the other sets in use. Firstly, annotations were done with canonical IPA anchors visible during the annotation process. In retrospect there is clear risk of biasing annotations towards the canonical forms under such a system, which was immediately clear in border cases where realization was close to the canonical form. This was particularly challenging for the contrasts of greatest interest—palatalized/velarized consonants—due in part to the absence of canonical examples for such sound contrasts in the annotation set-up. Similarity between such items also made ambiguous instances harder to decide objectively. Segment boundaries were deliberately placed as close to word-level as possible rather than phone level to avoid assigning nonconfident phone spans, while also chunking the longer audio utterances into lengths more reflective of the Irish training data. This introduced some uncertainty about what characters are aligning to canonical spans, and necessitated string alignment of the annotations to the canonical forms of the words during evaluation. Although some guidelines were informally decided at the outset, as understanding of the task evolved, so did the annotations, which on a limited evaluation set of 146 utterances leaves room for inconsistency.

There were design flaws in the annotation protocol worth bearing in mind, though it is also worth reflecting on what a lone annotator who is not a native speaker of the Irish can realistically hope to achieve in annotating the language for evaluation data. Although the utterances were reviewed numerous times, lack of partners to compare with to get a sense of inter-annotator agreement is a considerable limitation. Reliability testing provides a way to identify and measure where annotations of specific distinctions are less reliable (Artstein, 2017). It may make intuitive sense that cases of similar labels or difficult distinctions make annotations more unreliable, but it takes something to compare to in order to verify. A background as an Irish L2 speaker of modest ability makes such unreliability far more probable, particularly for more granular distinctions. Cole et al. (1994) shows agreement between labelers who did not

speak the language at a sobering 34.41% for the full label set without stripping any diacritics, even with linguistic training. They do show an increase to 41.36% when diacritics are stripped, but compared to the native speaker agreement at 73.79% under the same conditions, there is still a considerable gap. Together with a notable increase in agreement for segment boundaries the longer the segment, which holds whether the annotater speaks the language, the decision in the current work to focus on longer segments and on broader phonetic categories stripped of non-phonemic diacritics also serve a purpose for annotation. Collapsing Irish phones into the same forms as English data which might not cover the same range of realization is not a decision free of complications, however. In total, the scale of limitations here calls for caution in interpretting results: low annotation reliability for data means any evaluation based on it cannot offer strong support for positive or negative conclusions.

6.4.2 Cross-corpus comparability

Efforts were made to make annotation of evaluation data consistent with regard to conventions used for the Irish dataset, but this illuminated inconsistencies with the TIMIT set, as exemplified by the aforementioned incompatability between English model outputs for palatal/velar offglides vs the convension used for the Irish data to represent them as a single character. This misalignment between character sets points to a more general issue of comparability between different corpora. Performance between MDD papers varies widely, with some such as Wu et al. (2021) reporting far higher F scores at 80.98% than Zhang et al. (2022) at 55.3% or Phan et al. (2025) at 35.0% at character level evaluation. The latter two papers seek to overcome the general lack of L2 speech required to train models like the former, necessitating a patchwork of different data sources to train on.

The quantity of data used for fine-tuning is one factor worth considering on its own; many performant experiments such as Wu et al. (2021) utilize corpora such as CU_CHLOE (Helen Meng et al., 2007), containing 34.6 hours of richly annotated L2 data—more than enough to fine-tune, validate, and test on.[1] When one can evaluate with the same set the bulk of the training data comes from, you can be reasonably sure that the annotation schema is the same, the symbols mean roughly the same thing to all annotators, acoustic recording conditions are controlled, and that metrics using these datasets are more or less comparable. This cannot be taken for granted with a patchwork set up. This cross-corpus mismatch is noted in other domains such as Speech Emotion Recognition, as shown by Zong et al. (2022), which demonstrated improvements in employing transfer subspace learning to the problem.

Within a language, annotation schemas between corpora differ. They can prioritize different annotation targets, use different inventories, or record in different conditions. Between languages, one cannot assume that the same symbols mean the same thing. Identical symbol sets between languages are highlighted in Farooq and Hain (2022) *not* to be acoustically identical. They propose training mapping networks to make distributions from different model vocabularies comparable, demonstrating a resultant fusion of monolingual models with performance improvements compared to their component models in isolation. Another alternative to sidestep symbol mismatch might be to decompose the phones into features and train on those following Shahin et al. (2025). This could make the mapping between phone symbols and their language-specific realizations explicit, reducing spurious overlap in shared symbol inventories for segment-level MDD. Their work also notably increased contribution of monolingual data since it can train for features actually present in L2 speech with less

[1] Though it should be noted that some monolingual English data from TIMIT is used for fine-tuning as well.

direct exposure to it. Importantly, performance dropped when models were trained with dictionary form transcriptions over phonetic transcriptions, mirroring the performance difference between our TIMIT-trained English model and the Irish model trained on dictionary forms of words. For our case, collapsing the phone set did simplified annotation, but also increased the overlap between symbols that inarguably represented different things to different models, as exemplified by the velar offglide mismatch discussed above. A more systematic approach to ensuring phone set comparability like in the examples above might allow us to combine monolingual data in a more informed configuration.

6.5 Final Thoughts

Given the possibility that these experiments were implemented as intended and evaluated properly under the data constraints we were obliged to contend with, we cannot fully distinguish the efficacy of the methods from the ability of our evaluation to measure them. We concede that there are implementation limitations that could be improved for better performance, but also see reason for caution with respect to evaluation robustness. To reiterate, this is not to say that these methods would have succeeded under more charitable constraints, simply that the experiment as presented here cannot verify either way.

Because the structural limitations on the evaluation impose limits on whether an improvement can be cleanly detected, the most pressing priority for future experiments would be to establish a reliable evaluation infrastructure, taking into consideration reliability and comparability among datasets used. If that condition is fulfilled, we can consider technical extensions potentially worth pursuing.

7 Conclusions

In this paper we explored two avenues of overcoming data-sparcity for CAPT in the low-resource conditions faced by Irish. Neither the models trained on synthetically augmented data nor the multilingual ensembles provided a statistically significant improvement over baseline, with all versions of the ensemble significantly underperforming it.

Comparison of relative performance did reveal the utility of a light-weight trained selector block and entropy-based confidence measures for balancing ensembles, while specialist models and confidence pooling similarly helped, but overlapped in function. Data synthesis with English ASR transcriptions of Irish speech did better than adversarially paired versions of the same data, but enrichment of synthetic data with fuller variation of possible mispronunciation patterns was not explored here.

There are a number of limitations worth mentioning in this context. Foremost among them perhaps being annotation reliability and cross-corpus comparability. Annotation reliability was underdefined due to a lack of inter-annotator agreement and confounds due to annotator background. Corpora with different annotation schemas are not directly comparable even with overlapping phone sets. Experiments like ours seeking to leverage multiple data sources need to account for these cross-lingual differences.

In light of these caveats, we do not believe it is possible to draw strong conclusions about the presented results—neither positive nor negative. Though the methods used draw upon works that have shown more encouraging performance when applied under less challenging data constraints, there is no guarantee these approaches are suitable for MDD, but it may be worth testing them under more charitable constraints regardless. The results seen in the current work reiterate the need for a minimum of reliable evaluation data to adequately support MDD for low-resource languages.

8 Future work

The lack of readily-available data left us in the unenviable position of having no reliable data to evaluate with, and necessitated weaving together disjointed datasets with no clear way to compare them. Lack of evaluation data is a stubborn problem with no easy solutions, but overcoming the comparability problem can present several ways forward together with the methods explored here.

Cross-linguistic comparability was a particular thorn for our ensemble. A formal mapping for the disparate phone sets following Farooq and Hain (2022) would likely be the most direct solution for comparability between constituent models. Beyond this, overconfidence and imbalance between models was still an issue, and there are several refinements that could help. One comes directly from Laptev and Ginsburg (2023), specifically *tsallis entropy* as a basis for confidence calculation, which would give some control over the sensitivity of the entropy calculation with an *alpha* parameter. This measure becomes Gibbs entropy when alpha is 1, and acts similarly to temperature scaling when between 0 and 1. Another realistic extension going beyond Gitman et al. (2023) could be to train a more granular model selection block whereby phone outputs could be weighed individually for a model instead of each model being weight against eachother as a whole. This would necessitate more of the training data we try here to avoid, but could be worth consideration in the case where models show highly variable confidence values between different phones.

Enriching the training data to approximate mispronunciations does partially bridge the gap between separate inventories such that they share phone space, but the absence of a DPAD equivalent in our work, suggests a concrete extension to expand this coverage. Diversified output for an ASR approach analogous to our own might be achieved by means of shallow fusion of a Language Model (LM) together with Beam Search, whereby a scaling α value weighing the LM contribution might distort English ASR transcriptions of Irish speech to better adhere to English phonotactics, mimicking the sort of phonetic assimilation humans do when exposed to unfamiliar speech (For an example of L1 interference on L2 perception, see Kartushina and Frauenfelder, 2013). With the LM potentially acting to steer outputs preferentially towards English phonotactics, we could diversify the outputs via a Diverse Beam Search (Vijayakumar et al., 2018) similar to that used in Zhang et al. (2022). This expanded ASR approach might be preferable for low-resource languages like Irish where expertly annotated L1-L2 word pairs and L2 knowledge databases detailing common phone confusions are unavailable. Whether the combination of Irish speech through an English ASR with an English LM approximates realistic phone confusion pairs is an empirical question, but a concrete one worth exploring, and might be verified using higher resource languages for which such pairs are available.

Diversified synthetic speech that the DPAD analog generates may also be of use in a phonological feature-level MDD system following Shahin et al. (2025) as well. Since synthetic speech is highly determinative from input transcriptions, it might be more reliably broken down into precise phonological features to be used in such an MDD system. This might be preferable to using dictionary form transcriptions of speech, which normalizes variation to one form, and could increase coverage of L2-realistic mispronunciations more or less at will.

Another alternative might be to model the idea of L1 interference as phone-level code-switching, combining the multilingual nature of our ensemble idea, but sidestepping the relative confidence and alignment problems of training separate models by training a unified multilingual model. Code-switching multilingual models are trained with separate tokenizers for each language and perhaps synthetic examples to help define switching within a word. Feature-level decomposition could then be leveraged to ensure the different tokenizers adhere to their language specific realizations of each token.

finish train of thought to tie things together

9 Ethical Considerations

Data used for evaluation was drawn from FLEURS, and is governed by the Creative Commons License (CC-BY). The TIMIT dataset used for training is copyrighted by the Linguistic Data Consortium (LDC) and although the full version is available for $125 for academic entities, limited versions are available for free online via unofficial repositories. The one used for the current work was available for free on Kaggle.[1] The Irish data used for training was scraped from the online Dictionary and Language Library at teanglann.ie developed by Foras na Gaeilge in adherence with their scraping guidelines. I was unable to locate terms defining reuse, but reached out to Foras na Gaeilge to disclose this use and request clarification about their policies.

As stated earlier, speaking of Irish as one cohesive variant does not account for the distinctive variation between dialects. Here, we only considered Ulster Irish as the measure of "correct" Irish, but this is not a neutral choice. Within this dialect, we do not either consider a natural variation of realized pronunciation, focusing only on one form of pronunciation taken as canonical. Both of these assumptions do not reflect the linguistic reality of Irish, and were simplifications intended to fascillitate the systems explored here as research prototypes, *not* prescriptive authorities of the language.

Though it perhaps goes without saying given lackluster performance, the experiments explored here are in no way ready for deployment in any pedagogical context. Rates of false rejection indicate any system built on these experiments as-is would likely be highly inconsistent, demotivating, and misleading. Rates of false acceptance further indicate systems of potentially questionable pedagogical value. Further validation and experimentation are necessary before trust in these systems can be earned.

[1] https://www.kaggle.com/datasets/mfekadu/darpa-timit-acousticphonetic-continuous-speech

10 AI Tools

AI chatbots ChatGPT, Claude, and Gemini were used in discussing ideas, summarizing work from a range of sources, providing leads for promising threads of inquiry, as well as suggesting options for overcoming hurdles. When coding, I called upon them to link me to tutorials or provide example code I could follow along with to enhance my understanding of challenging technical concepts. Some valuable insights I might have otherwise missed were illuminated by discussions with humans and chatbots, and I think my work was better for their input.

The agentic coding system Claude Code was used in a number of productive ways. It helped threading experiments together and scaling them into to a coherent evaluation, debugging by providing code to test my systems were behaving as I expected them to, and in generating illuminating graphics I could use to check my intuitions and show my work. Learning to use Claude Code let me shift my focus from time-consuming implementational details to planning and testing, and I think that my work was made better by its assistance.

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